

• AVAILABLE FOR BACKEND WORK

## Arash MohammadGholinejad

Backend Developer — C#/.NET & Python

# Backend systems in two languages, one set of habits.

I like hard problems, I learn fast, and most of what I build on my own ends up open source.

See projects

Get in touch

Email   GitHub   LinkedIn   Résumé

C# / ASP.NET Core

REQUEST →

Web API — controllers, JWT auth

Application — MediatR (CQRS)

Domain — entities, invariants

Infrastructure — EF Core → SQL Server

Python / Django

REQUEST →

DRF views — JWT auth, serializers

Services — business logic

Models — Django ORM

Data — PostgreSQL + Redis cache

// about

# From ML coursework to production backends

I studied Computer Engineering and got into machine learning and deep learning along the way, through coursework and a lot of self-study. Backend development is where I ended up, because it's the work I actually enjoy doing every day: figuring out how a system should fit together and then making it hold up.

**What keeps me interested is the hard part.** I like problems without an obvious answer, and I'll stay with one until I understand it properly, not just until the error goes away. I learn quickly when a project needs something I don't know yet. Outside of work I keep my projects public on GitHub from the first commit, and when I find a bug in an open-source tool I use, I send the fix to the original project instead of only patching my own copy.

// where I've worked

# Experience

- Software Engineer** **Metable**

Dec 21, 2024 - Sep 22, 2026

  - Developed and maintained the company's software products.
- Software Engineer** **Hamkaran System**

Jul 24, 2024 - Dec 20, 2024

  - Developed and maintained enterprise ERP applications on the in-house Rahkaran framework.
  - Worked within existing enterprise systems, implementing and maintaining business functionality.

C# · ASP.NET
- Software Engineer** **Alborz Afraz Tabarestan (Dornica)**

Jul 23, 2023 - Jun 20, 2024

  - Developed and maintained backend services and RESTful APIs.
  - Designed and implemented an API Gateway service for centralized API management, routing and integration across services.
  - Built an online ArcGIS validation service and ArcGIS plugins for automated data/config validation and GIS workflows.
  - Refactored and modernized multiple projects for architecture, performance and maintainability.

Python · FastAPI · Celery
- Software Engineer** **Zarebin (Hamrah Aval Internship)**

Oct 2022 - Feb 2023

  - Developed and maintained backend services.
  - Designed and implemented RESTful APIs and managed database operations.

Python · Django
- Software Engineer** **Jadooye Fekr (Internship)**

Jun 2020 - Sep 2020

C# · ASP.NET · Web Forms

// toolbox

# Skills

## .NET / C#

- ASP.NET Core Web API
- MVC
- Entity Framework Core
- LINQ
- MediatR
- AutoMapper
- xUnit

## Python

- FastAPI
- Django
- Flask
- DRF
- SQLAlchemy
- Pytest

## Databases & caching

- SQL Server
- PostgreSQL
- MongoDB
- Redis

## Tools & DevOps

- Docker
- Git
- GitHub Actions
- Swagger / OpenAPI

## Architecture

- Clean Architecture
- DDD
- CQRS
- API Gateway
- Microservices

## Practices

- OOP
- SOLID
- Design Patterns
- TDD
- Refactoring



// selected work

# Projects

Four projects that show the same habits — layered architecture, JWT auth, Docker, CI — applied on both stacks.

GET /catalog/movies

200 OK

## IMDB — movie & series API

IMDb-style catalog API on ASP.NET Core 8, built around Clean Architecture and tactical DDD so the domain enforces its own rules — aggregates and invariants — instead of leaning on validation at the API edge. CQRS via MediatR keeps reads and writes from tangling as the surface grew to 45 operations.

C#

ASP.NET Core

MediatR

EF Core

SQL Server

xUnit

Docker

[View code on GitHub →](#)

POST /shop/checkout

201 Created

## ProductKrait — e-commerce platform

Django e-commerce build where speed of delivery was the actual constraint, so the architecture stays straightforward on purpose: catalog, cart, orders and OTP-verified payments on Django's own ORM and auth conventions, with a JWT-secured REST API layered on top rather than replacing the server-rendered admin/staff dashboard.

Python

Django

DRF

JWT

Docker

[View code on GitHub →](#)

[Live demo →](#)

GET /feed/following

200 OK

## AdderHub — social network

Instagram-style social app built with Django, where the interesting problem wasn't the feature list but serving it twice — a server-rendered site and a JWT-secured REST API — without duplicating the rules for what belongs in a feed. Redis sits in front of the one query that's read far more than it's written.

Python

Django

DRF

Redis

Docker

[View code on GitHub →](#)

[Live demo →](#)

## Store Management — retail API

Store REST API on ASP.NET Core 8, layered into Domain/Application/Infrastructure/Persistence so "can this order be placed" is a rule the domain checks once, not logic repeated in every place stock gets touched — orders are validated against stock and saved atomically, not checked-then-hoped-for.

C#

ASP.NET Core

EF Core

SQL Server

AutoMapper

Docker

[View code on GitHub →](#)

Auth Service (MongoDB)

JWT auth & user management on ASP.NET Core + MongoDB, PBKDF2 hashing, owner-or-admin authorization, Problem Details errors.

[Code →](#)

API Gateway &amp; ArcGIS validation

Built at Dornica: centralized API routing/gateway and an automated ArcGIS config-validation service. Closed-source client work — described here, code not public.

[Details →](#)

```
// how it's built
```

## Two projects, closer look

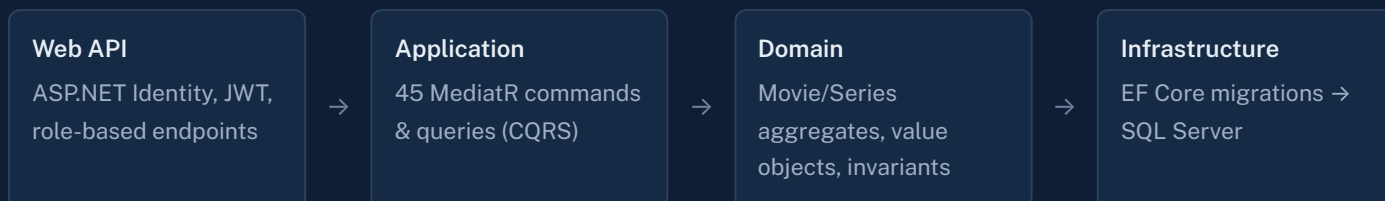
The résumé says what these do. This is closer to how — the layers, the calls I made on the way, and the pattern behind them.

# IMDB — proving the domain layer earns its keep

Most of my day job is CRUD over an existing schema. IMDB was the project where I wanted a domain complex enough that Clean Architecture and DDD were the right tool, not decoration — a catalog where "what's a valid movie" is actually a rule, not just a database constraint.

[View code on GitHub →](#)

## ARCHITECTURE



## DECISIONS

### CQRS over one service per controller

Splitting reads and writes into MediatR handlers kept 45 operations organized as the API grew, instead of a handful of bloated services.

### Invariants on the aggregate, not the API

Validation lives inside the entities, so an illegal state is impossible to construct — not just rejected at the edge by a filter.

### Tests aimed at behaviour, not endpoints

xUnit tests exercise domain rules and handlers directly, so a broken invariant fails fast without spinning up the whole API to notice.

## THE PATTERN

*A simplified version of the invariant-enforcing entities used across IMDB's aggregates:*

```
public sealed class Movie : AggregateRoot
{
    private readonly List<CastMember> _cast = new();

    public static Result<Movie> Create(
        string title, DateOnly releaseDate, Rating rating)
    {
        if (releaseDate > DateOnly.FromDateTime(DateTime.UtcNow))
            return Result<Movie>.Fail(
                "Release date cannot be in the future.");

        return Result<Movie>.Ok(new Movie(title, releaseDate, rating));
    }

    public void AddCastMember(CastMember member)
    {
        if (_cast.Any(c => c.PersonId == member.PersonId))
            throw new DomainException(
                "This person is already cast in the movie.");
    }
}
```

```
        _cast.Add(member);  
        RaiseDomainEvent(new CastMemberAddedEvent(Id, member.PersonId));  
    }  
}
```

# AdderHub — one codebase, two front doors

A social app usually needs a website and an API. AdderHub serves both — server-rendered pages for the site, a JWT-secured DRF API for anything else — from the same models and the same business logic, instead of maintaining two parallel implementations of "what's in my feed."

[View code on GitHub →](#)

## ARCHITECTURE

### Server-rendered views

Django templates, sessions

### DRF API

JWT auth, Swagger/ReDoc

↓ both call the same service layer ↓

### Services

Feed, posts, follows — one place decides the rules

↓

### Data

Django ORM → database, Redis in front of the feed query

## DECISIONS

### One codebase, two front doors

The website and the API call the same service functions, so a rule about the feed only has to be right once.

### Redis in front of the feed, not the whole app

The following/explore feed is the most repeated read by far, so it's the one thing cached — not a blanket cache over everything.

### UUID keys on public-facing models

Post and profile IDs don't leak how many users or posts exist, without touching anything else in the schema.

## THE PATTERN

*A simplified version of the shared service layer behind both front doors:*

```
class FeedService:
    """Shared by the server-rendered view and the DRF
    endpoint -- one place decides what's in a feed."""

    def __init__(self, cache):
        self.cache = cache

    def get_following_feed(self, user, page=1):
        key = f"feed:{user.id}:{page}"
```

```
if cached := self.cache.get(key):
    return cached

posts = (
    Post.objects
    .filter(author__in=user.following.all())
    .select_related("author")
    .prefetch_related("likes", "comments")
    .order_by("-created_at")
)
page_obj = paginate(posts, page)
self.cache.set(key, page_obj, timeout=60)
return page_obj
```

// open source

## Building in the open

Every personal project above is public on GitHub from its first commit, not cleaned up and published after the fact — README, CI and all. Where I'm using someone else's project, I'd rather send a fix upstream than keep a private patch.

Beyond backend work, I maintain two forks of **DS4Windows** with pull requests sent upstream.

↑ **DS4Windows (fork) — PR open**  
Fixed a PresetIntroText localization key casing mismatch; contributing upstream fixes to a community-maintained controller driver.  
[github.com/codesbygom/DS4Windows5](https://github.com/codesbygom/DS4Windows5) →

↑ **DS4Windows (second fork)**  
A second, earlier fork of the same project, also carrying a pull request back upstream.  
[github.com/codesbygom/DS4Windows-1](https://github.com/codesbygom/DS4Windows-1) →

// background

# Education

## B.S. in Computer Engineering

Babol Noshirvani University of Technology

2016 - 2021

Served as a teaching assistant for programming courses.

Machine Learning Specialization — Coursera, Feb 2024

Deep Learning Specialization — Coursera, Mar 2024

## Let's talk about your backend.

Open to remote backend roles and freelance work in .NET or Python — API design, Clean Architecture refactors, or building something new from a spec.

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GITHUB

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Arash MohammadGholinejad — Backend Developer

Open to remote and freelance work.